

Hardware-Efficient Neural Architecture Search for Wireless Signal Classification on Edge Devices

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3,539 params

Ultra-compact CNN (<20k target)

90.1% accuracy

Real over-the-air test data

13.8 KB model

~3.5 KB with INT8

Sub-ms inference

On iPhone Neural Engine

1 MOTIVATION & PROBLEM

- Growing EM complexity demands automatic signal classification.
- Manual feature engineering doesn't scale; deep models exceed edge budgets.
- Applications: spectrum monitoring, cognitive radio, interference detection.
- Goal:** architectures that are *both* accurate *and* deployable.

3-Class Wireless Technology Identification



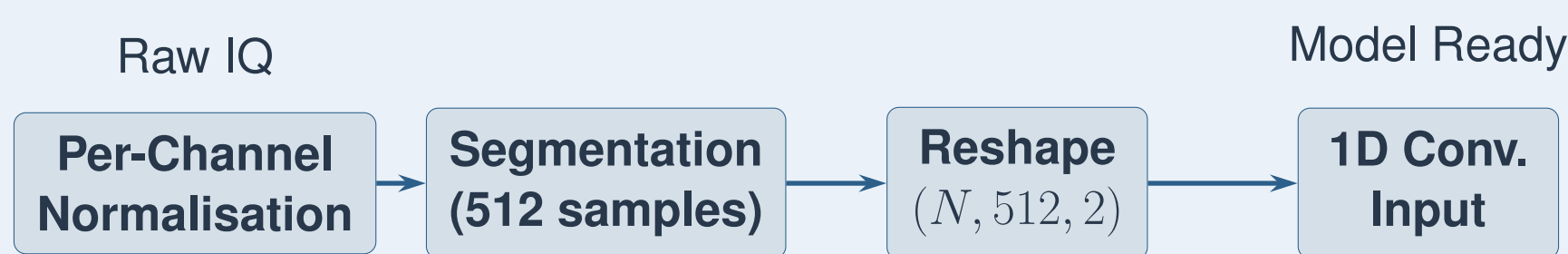
512-sample I/Q segments → CNN classifier

2 DATASET & PREPROCESSING

Fontaine & Shahid — Over-the-Air IQ Signals

Property	Value
Acquisition	USRP, 1 Msps, OTA (Gent, BE)
Total files	135 (45 per class)
Classes	LTE, DVB-T, WiFi
Segment length	512 IQ samples
Input shape	512 × 2
Format	Binary IQ files

Preprocessing Pipeline:



3 HARDWARE DESIGN CONSTRAINTS

Parameters <20,000

Model Size (FP32) <80 KB

Inference Latency <1 ms

Streaming-Friendly Structure Branch-free

Goal: Meet every constraint above while maximizing classification accuracy

4 STATE-OF-THE-ART COMPARISON

Benchmarking Against Recent Models (OTA & Synthetic)

Model	Params	Data	Accuracy
NAS (ours)	3,539	OTA	90.1%
Ide et al. (2024)	~138,000	SYN	96.8%
Shebert et al. (2021)	2.1M–67.1M	SYN	94.5% (SNR>0dB)
Zhang et al. (2021)	~165,000	OTA/SYN	96.2% (OTA)
Fontaine et al. (2019)	213,000/6,400	OTA	97.8%/88.0%
Gu et al. (2019)	99,900	OTA	95% (high SNR)

Smallest model among comparable OTA-validated methods — a fraction of typical parameter counts

5 NAS METHODOLOGY

Hardware-Aware Evolutionary Search

Framework: Evolutionary (Tournament + Crossover + Mutation) • Candidates: 160 architectures

Search Space	Search Operators
1D Conv: std. & separable - Optional LSTM blocks - Compact dense layers - Global average pooling - Kernel size: 5 (found) 2 conv. + dense head - Branch-free, sequential	- Tournament selection - Uniform gene crossover - Mutation rate: 10% - Elitist replacement - Population: 16 models 10 generations, 160 evals ~2h on Apple M3 CPU

Fitness Function: $F = 1.2 \cdot (1 - A) + 0.25 \cdot P_{\text{penalty}}$

Accuracy weight 1.2

Compactness penalty 0.25

6 NAS-DISCOVERED ARCHITECTURE

3,539 Parameters — ~13.8 KB (FP32)

Layer	Type	Params	Memory
Input	InputLayer	0	(512, 2) I/Q
Conv1	Conv1D ($k=5$)	176	16 filters
BN+Drop	BatchNorm	64	Stabilisation
Conv2	Conv1D ($k=5$)	2,592	32 filters
BN+Drop	BatchNorm	128	Stabilisation
GAP	GlobalAvgPool	0	Dim. reduction
Dense	Dense	528	16 units
Output	Dense	51	3 classes, softmax
Total		3,539	~13.8 KB

1.41M

Total MACs
per inference

~3.5 KB

INT8 Footprint
HW-dependent

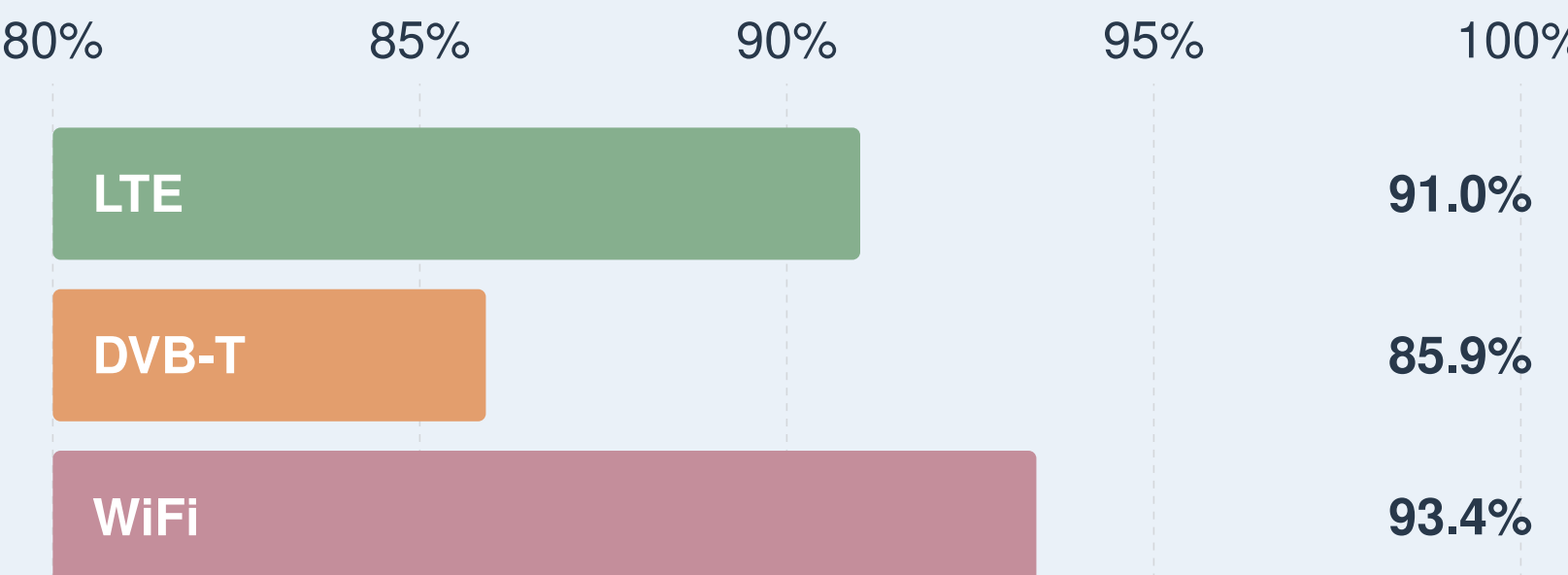
225 KB

Artix-7 BRAM
Ample headroom

7 PERFORMANCE RESULTS

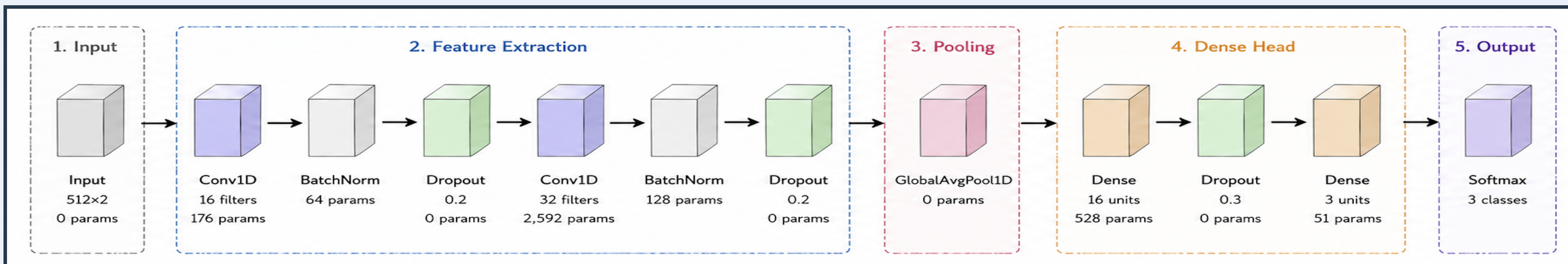
Metric	Value
Train Accuracy	94.5%
Validation Accuracy	87.7%
Test Accuracy	90.1%
Macro Precision	90.2%
Macro Recall	90.1%
Macro F1-Score	90.1%

Per-Class F1-Score — Test Set:



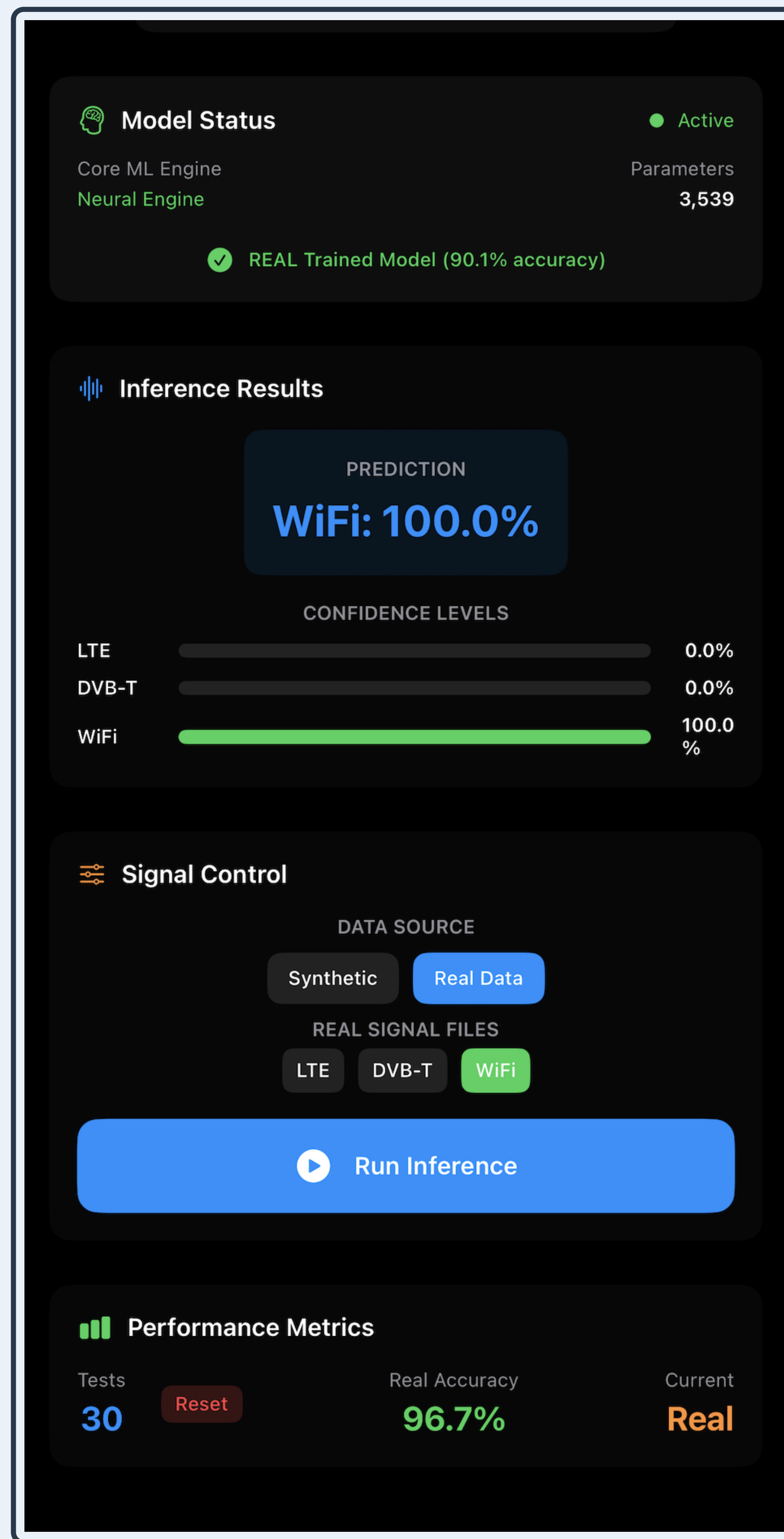
Main confusion: DVB-T ↔ WiFi, due to spectral similarity between OFDM-based technologies at baseband

NAS-DISCOVERED ARCHITECTURE — Layer-by-Layer



8 REAL-WORLD DEPLOYMENT

iPhone 17 Pro Max — On-Device Validation

Note: Inference runs on pre-recorded IQ segments from the dataset, not on live over-the-air samples.

9 DEPLOYMENT PIPELINE & HARDWARE

Sub-ms
Neural Engine Inferencemlprogram
Core ML Format0% Cloud
Fully On-DeviceA19 Pro
SoC + Neural Engine

Headroom vs. Design Budget:

Parameters 3,539 of 20,000
82% headroom leftModel Size (FP32) 13.8 of 80 KB
83% headroom left

Constraint	Target	Achieved
Inference Latency	<1 ms	Sub-ms
Cloud Dependency	None	None

Detailed on-device latency and power characterization are left as immediate future work

10 CONCLUSIONS & FUTURE WORK

3,539
Total Parameters
<20k target met90.1%
Test Accuracy
Real OTA data13.8 KB
Model Size (FP32)
~3.5 KB INT8Sub-ms
Inference Latency
iPhone Neural Engine

Live IQ Acquisition

RF Front-End Integration

Latency &
Power ProfilingFPGA / ASIC
ImplementationEvolutionary NAS discovers architectures that strongly favor **efficiency** while preserving useful **accuracy**, enabling spectrum monitoring under strict edge-device constraints.

Supported by EU NextGenerationEU — Red.es

PID2022-138078OB-I00 • PDC2023-145808-I00

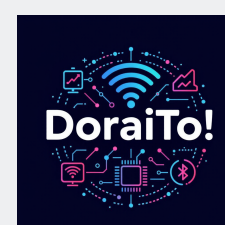
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Financiado por la Unión Europea

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